

Viscoelasticity

- Mechanical behavior of polymers to applied stress or strain depends on or of loading
- – at low strain-rate, obey *Hooke's law*
i.e. $\sigma = E\varepsilon$
- – at low strain-rate, obey *Newton's law*
i.e. $\sigma = \eta d\varepsilon/dt$
- Most polymers are in between viscous liquids and elastic materials (viscoelastic)
- Viscoelasticity is highly dependent on temperature.

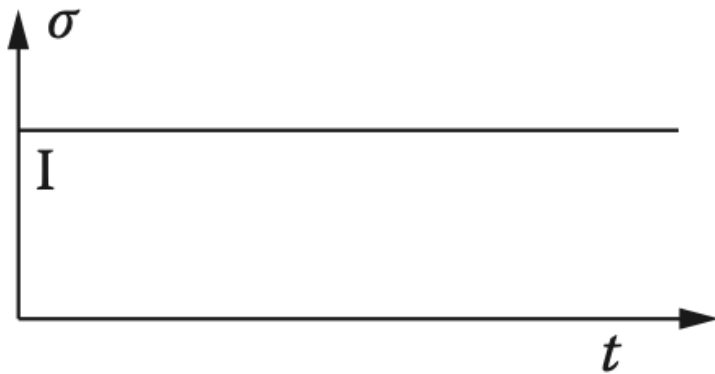
why (molecular level)?

Viscoelasticity

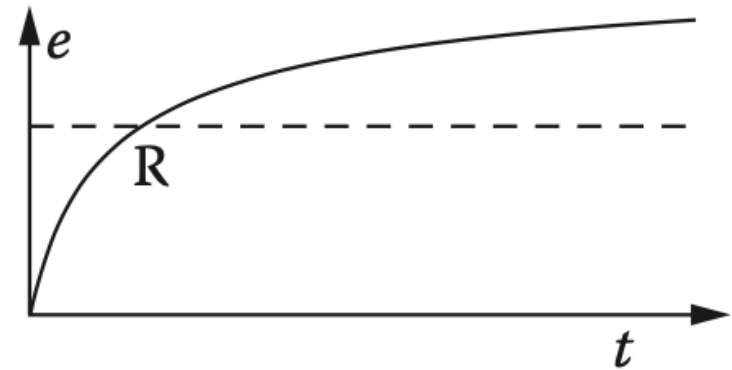
- What happens when weight is suspended on a polymer filament?
 - Strain increases over time due to stress-induced molecular rearrangement in the solid. Upon release, the molecules slowly recover to original state
 - This is called – a manifestation of viscoelasticity.
- How then can we use polymers in engineering?
 - Polymers do not creep and they fully recover on removing the stress
- Thus is important in the use of polymer as an engineering material.

Viscoelastic responses: creep

Stress system

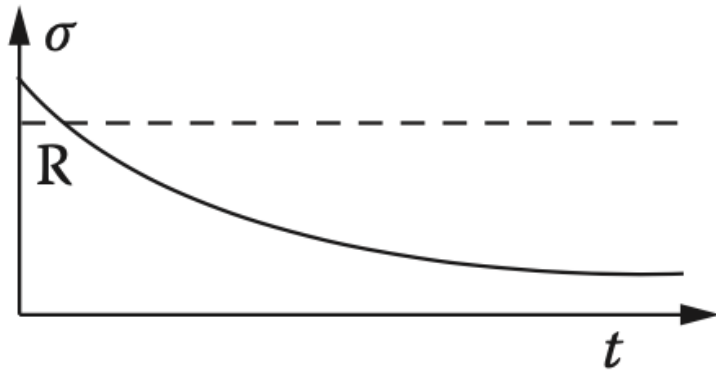


Strain system

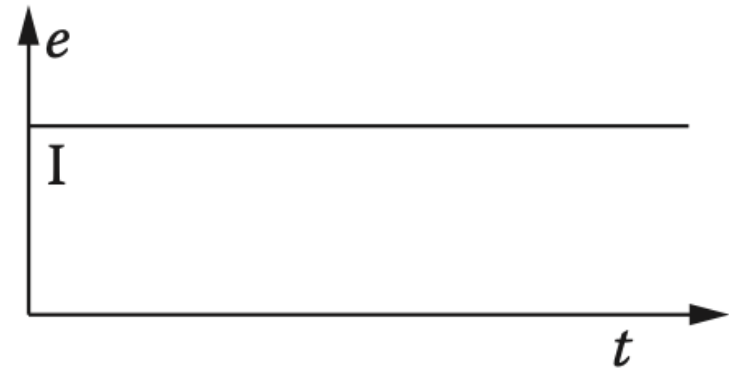


Viscoelastic responses: relaxation

Stress system

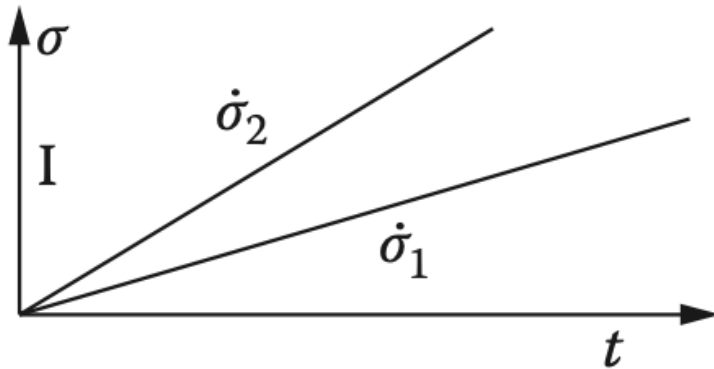


Strain system

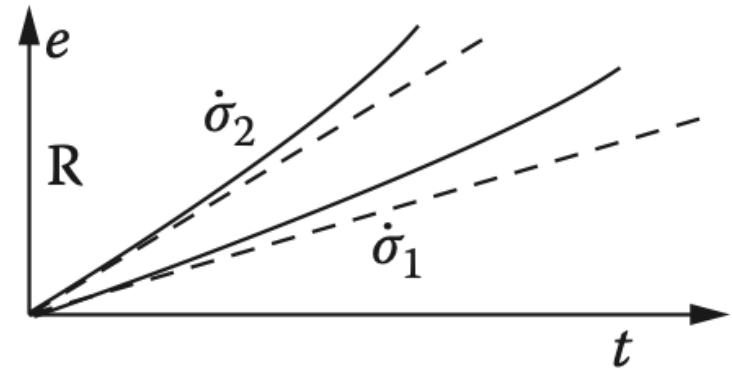


Viscoelastic responses: constant stress-rate

Stress system

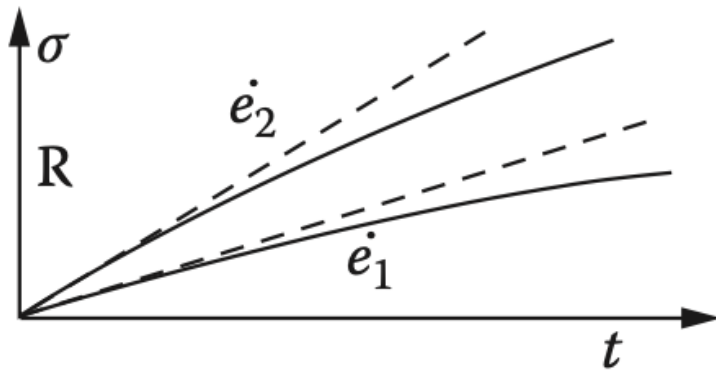


Strain system

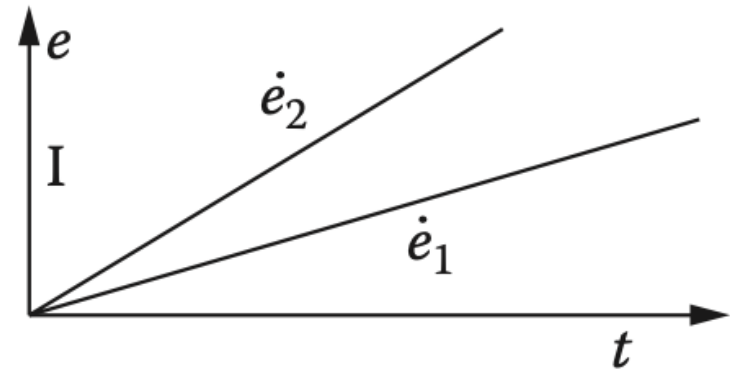


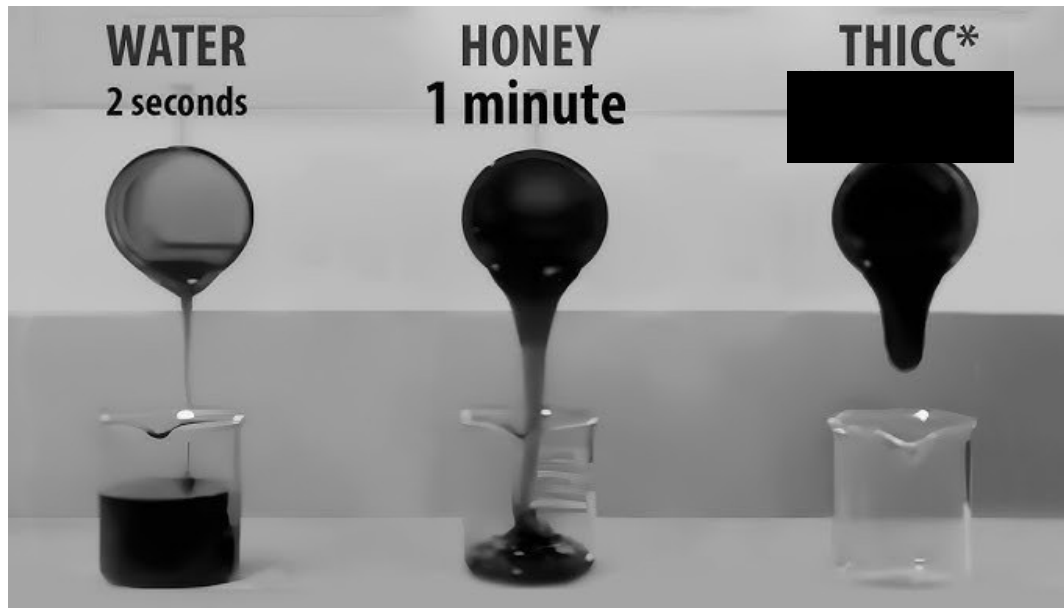
Viscoelastic responses: constant stress-rate

Stress system



Strain system



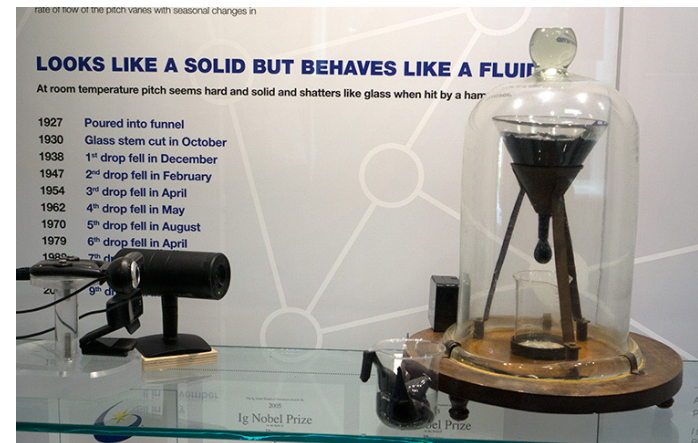


How long does it take for 1 drop of pitch to form & release?

GROUP ACTIVITY!

Extremes of viscoelasticity: pitch

- Pitch: extremely viscous (polymer)
- Produced through distillation of carbonaceous materials
- Rosin: derive from plants (usually pine trees)
- Bitumen/asphalt: derive from petroleum

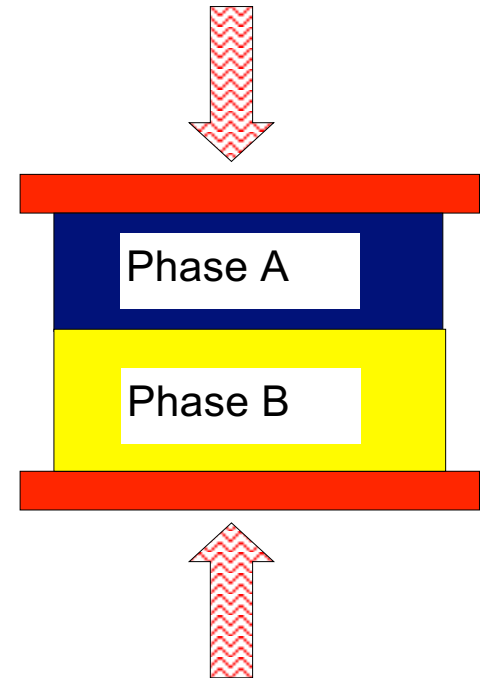


https://en.wikipedia.org/wiki/Pitch_drop_experiment

<https://vimeo.com/event/4721898>

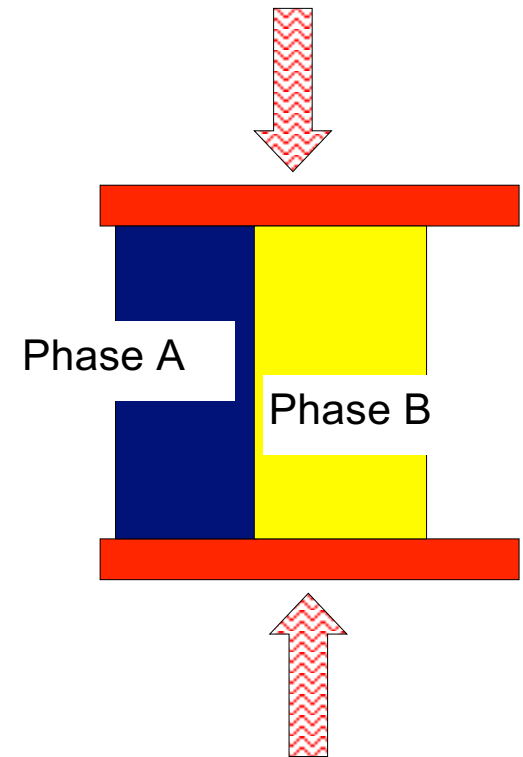
Isostress

- Imagine parallel slabs of material between platens that apply a load.
- Phase A has volume fraction V_A and modulus E_A ; Phase B has volume fraction V_B and modulus E_B .
- What is the composite modulus, E_C ?
- We assume because each phase sees the same stress (assuming same cross-sectional area).
- The stress, $\sigma = \sigma_C$, is therefore the field and the strain is the response (and the compliance is the property).



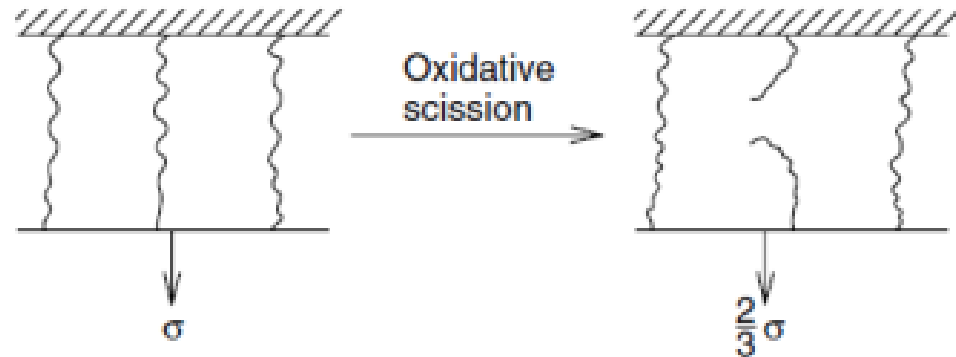
Isostrain

- Imagine parallel slabs of material between platens that apply a load.
- Phase A has volume fraction V_A and modulus E_A ; Phase B has volume fraction V_B and modulus E_B .
- What is the composite modulus, E_C ?
- We assume because each phase sees the same change in length.
- The strain, $\varepsilon = \varepsilon_c$, is therefore the field and the stress is the response (and the stiffness is the property).

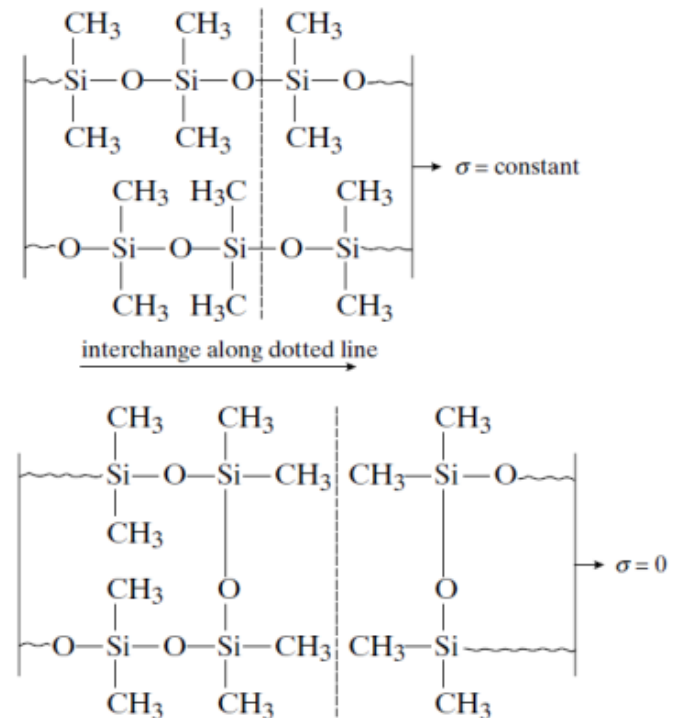


Molecular basis stress relaxation

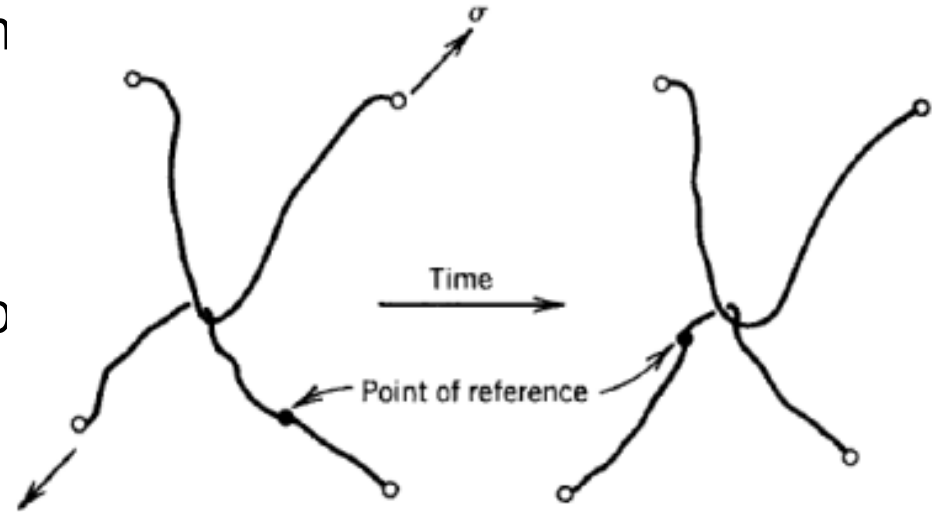
- : breakage of polymeric chains – reduces mechanical strength but increases the elongation during creep



- : chains exchange bonds leading to chain re-organization
 - Common in polysiloxanes and polyesters



- **Molecular relaxation**: energy dissipation through molecular vibrations (energy dissipation)
- **Thirion relaxation**: entangled chains slip *reversibly* over each other
 - Specific to polymers
- **Viscous flow**: polymer chains slip past each other under flow.
 - Permanent
 - More prominent in polymers



But how can we model/understand relaxation and creep?

- Two approaches: stress relaxation or strain relaxation

Understanding stress relaxation & creep

Maxwell model (isostress)

- : a mechanical device that resists motion through viscous friction e.g. used in hydraulic doors
 - The mechanical behavior is described as;

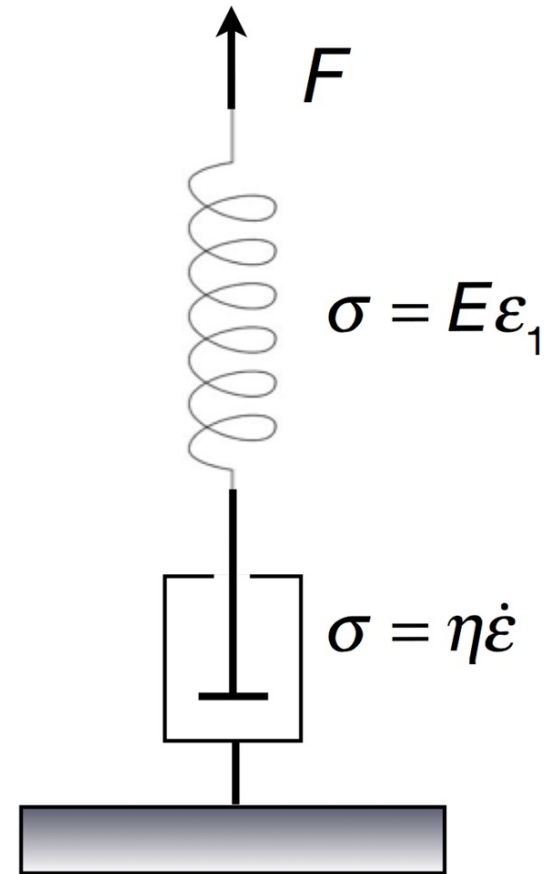
$$\sigma = \eta \frac{d\varepsilon}{dt} = \eta \dot{\varepsilon}$$

- : Elastic behavior follows Hooke's law, response to stress is;

$$\sigma = E\varepsilon$$

- Maxwell model places a spring and dashpot in series. Stress is same across the assembly, but strain is ht, hence;

$$\frac{d\varepsilon}{dt} = \frac{d\varepsilon}{dt} + \frac{d\varepsilon_1}{dt} = \frac{1}{E} \frac{d\sigma}{dt} + \frac{\sigma}{\eta}$$



Understanding stress relaxation & creep

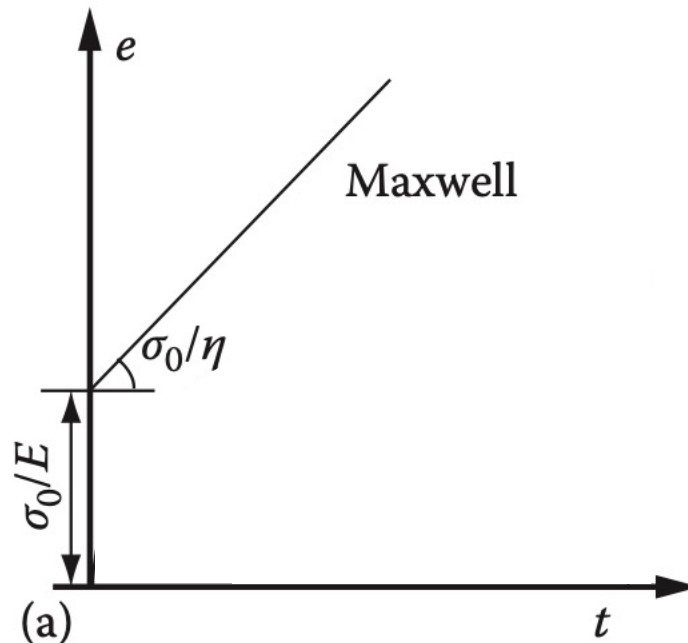
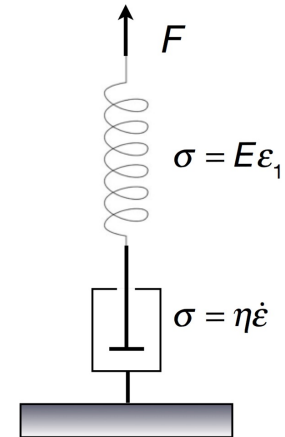
Maxwell model

- Creep = constant stress:

$$\sigma = \sigma_0$$
$$\frac{d\sigma}{dt} = 0$$

Therefore

$$\frac{de}{dt} = \frac{\sigma_0}{\eta}$$



Understanding stress relaxation & creep

Maxwell model

- Stress relaxation = constant strain

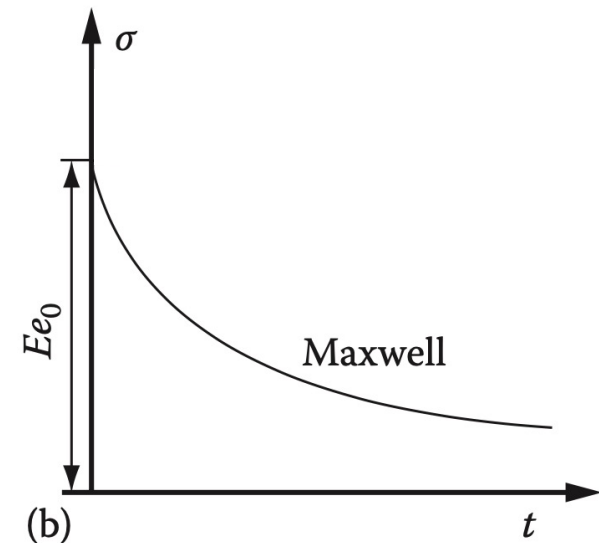
$$e = e_0$$

$$\frac{de}{dt} = \frac{1}{E} \frac{d\sigma}{dt} + \frac{\sigma}{\eta} = 0$$

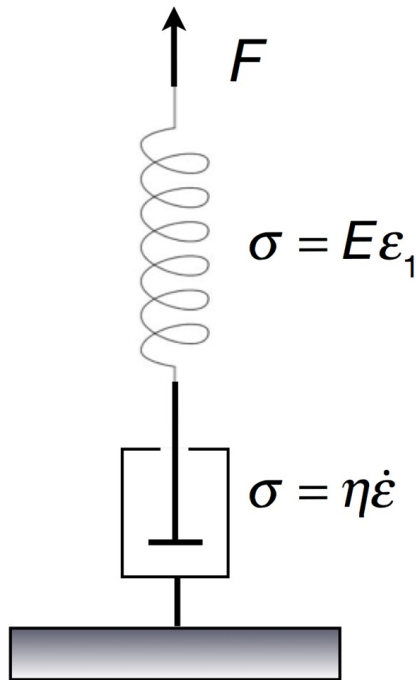
- Then;

$$\frac{1}{E} \frac{d\sigma}{dt} = -\frac{\sigma}{\eta} \Rightarrow \frac{d\sigma}{\sigma} = -\frac{E}{\eta} dt$$

$$\sigma = \sigma_0 e^{-(E/\eta)t}$$



Stress relaxation & creep: Maxwell model



- Can we estimate the relaxation time?
- Will the stress eventually dissipate and system relaxes back?

- Relaxation time, τ , is defined by;

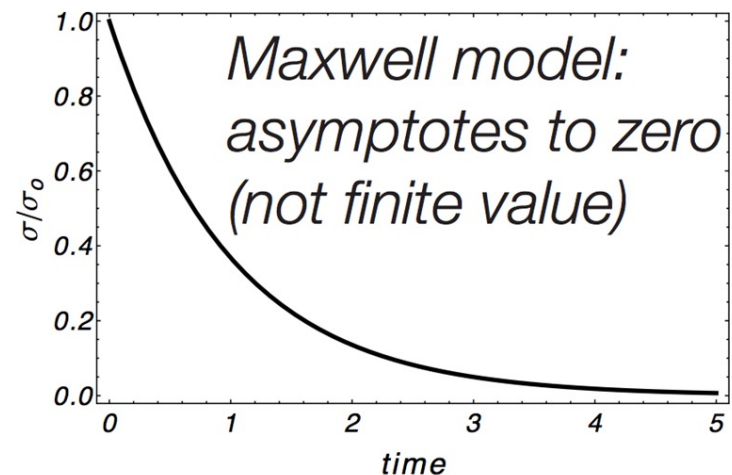
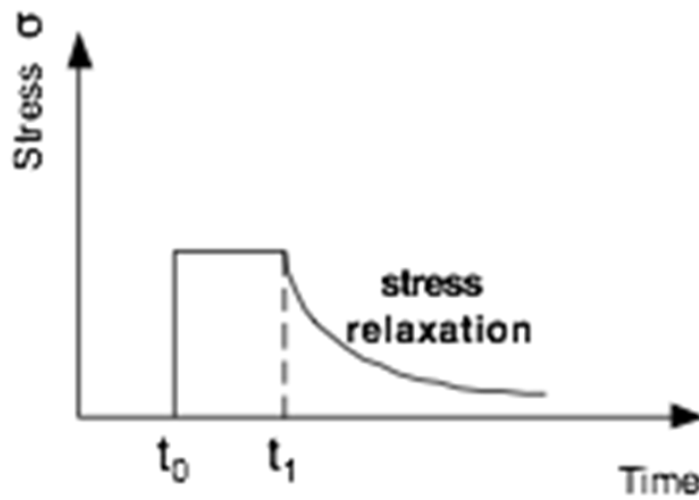
$$\tau = \frac{\eta}{E}$$

- Hence;

$$\sigma = \sigma_0 e^{-(E/\eta)t}$$

becomes

$$\sigma = \sigma_0 e^{-t/\tau}$$



Stress relaxation & creep

Kelvin/Voigt Model (isostrain)

Creep

- Strain is the same on each element, but stress is different
- At constant load;

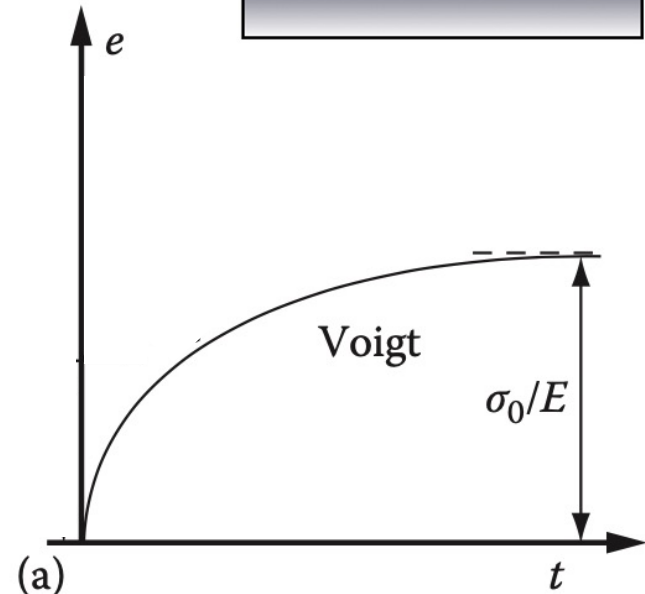
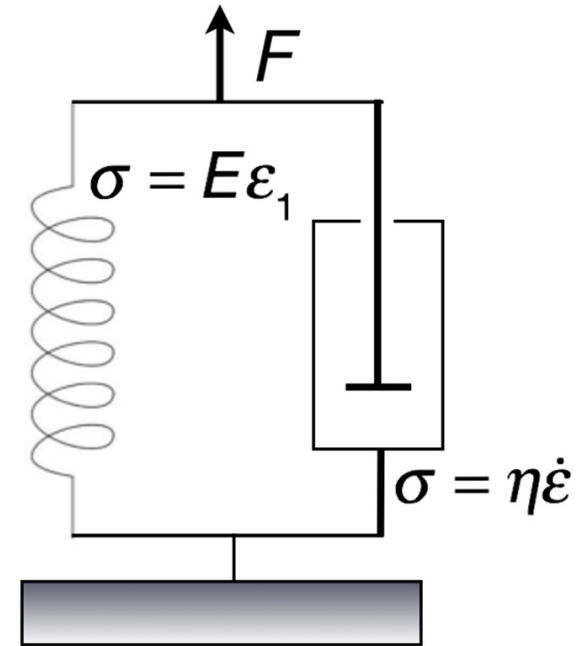
$$\frac{d\varepsilon}{dt} = \frac{\sigma}{\eta} - \frac{E\varepsilon}{\eta}$$

- Hence;

$$\frac{d\varepsilon}{dt} + \frac{E\varepsilon}{\eta} = \frac{\sigma_o}{\eta}$$

- Similarly, solving this equation gives us;

$$\varepsilon = \left(\frac{\sigma_o}{E}\right) \left[1 - e^{-Et/\eta}\right] = \left(\frac{\sigma_o}{E}\right) \left[1 - e^{-t/\tau}\right]$$



Stress relaxation & creep

Kelvin/Voigt Model (isostrain)

Stress relaxation

- Strain is the same on each element, but stress is different
- At constant strain:

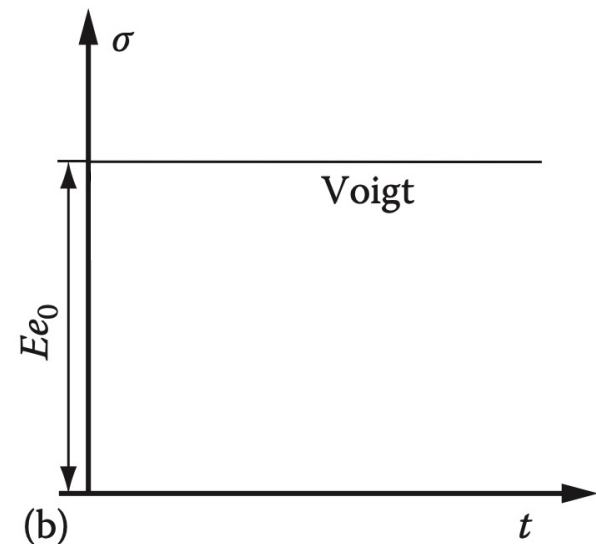
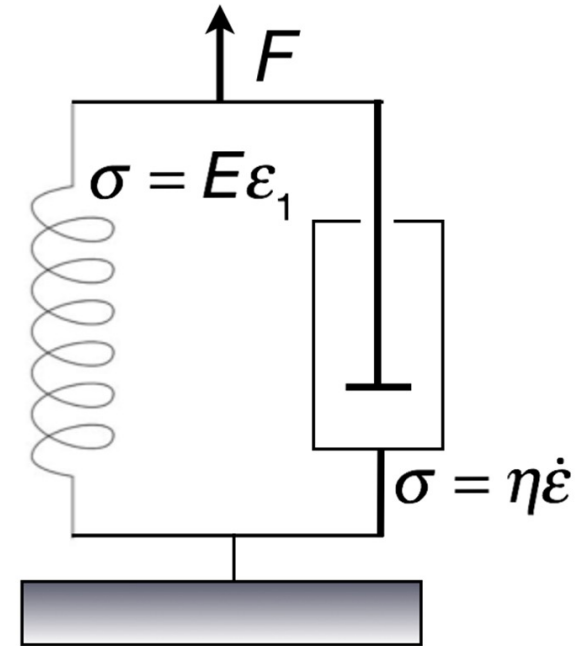
$$\frac{de}{dt} = 0$$

- Hence;

$$\frac{\sigma}{\eta} = \frac{E e_0}{\eta}$$

and

$$\sigma = E e_0$$



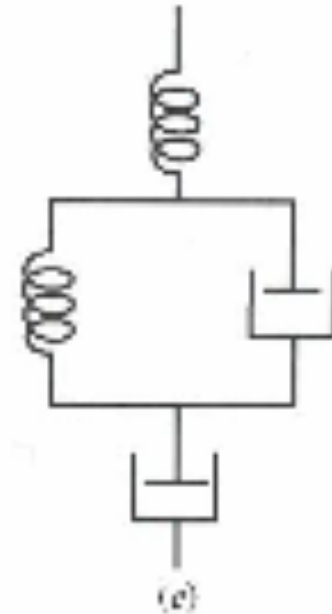
Comments

Neither the Maxwell or Kelvin-Voigt models work perfectly.

One works for creep and one for stress relaxation.

More complex models can be constructed based on various combinations of springs and dashpots.

This is a standard way of building accurate models of viscoelasticity.



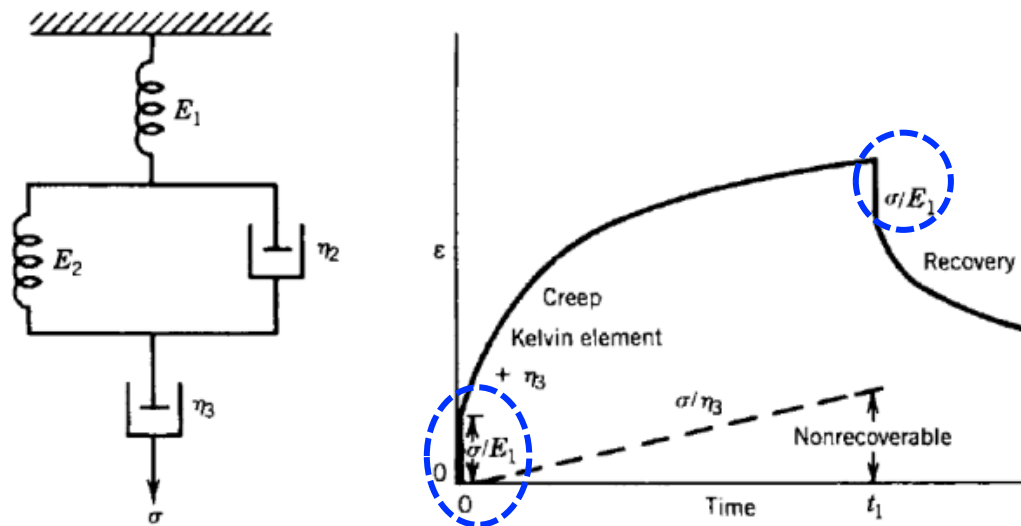
a 4-element model

Stress relaxation & creep

Burgers Model

- Four element models – combination of Maxwell and Kelvin models
 - At $t = 0$, $\varepsilon = \sigma/E_1$ - deforms right away and is first element to relax.
 - Also, $\varepsilon_3 = \dot{\varepsilon}\Delta t = \frac{\sigma}{\eta_3}\Delta t$ – this is non-recoverable strain
 - Strain on the Kelvin elements is; $\varepsilon_2 = \varepsilon - (\varepsilon_1 + \varepsilon_3)$
 - Hence, overall strain can be expressed as;

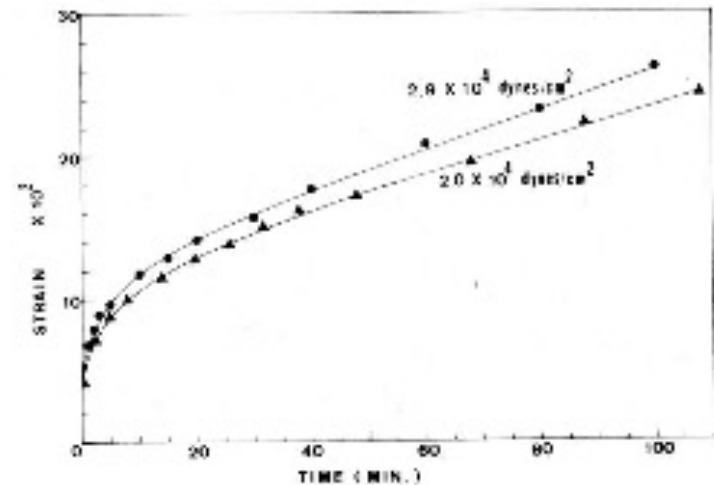
$$\varepsilon = \frac{\sigma}{E_1} + \frac{\sigma}{E_2} \left(1 - e^{-t/\tau_2}\right) + \frac{\sigma}{\eta_3} t$$



Example 1: Velveeta[®] cheese

Researchers made measurements on Velveeta[®] cheese:

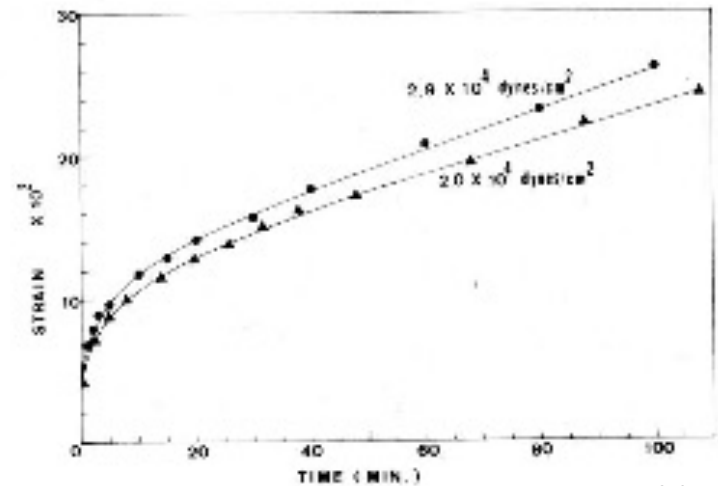
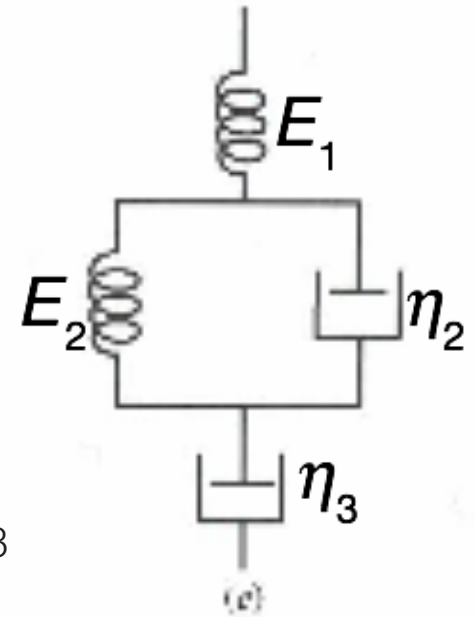
- 15 cm block with dimensions 4x6 cm
- compressive load of 4.9 N for 2 hours
- height of block measured every 5 minutes
- a second experiment used a load of 6.85 N
- the data are the symbols:



Example 1: Velveeta[®] cheese

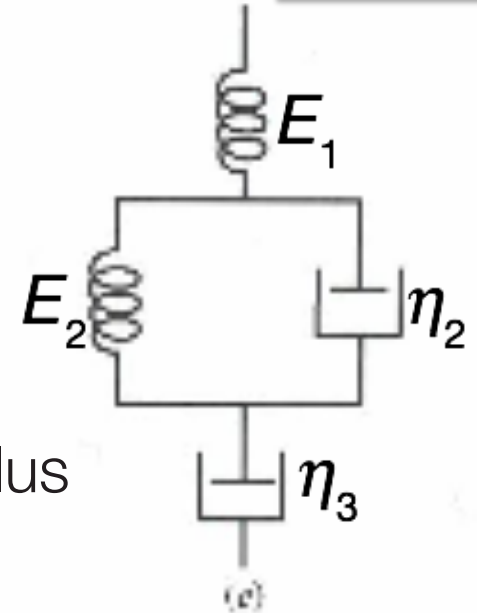
$$\varepsilon = \frac{\sigma}{E_1} + \frac{\sigma}{E_2} \left(1 - e^{-t/\tau_2}\right) + \frac{\sigma}{\eta_3} t$$

- at long times, $\varepsilon \rightarrow \frac{\sigma}{E_1} + \frac{\sigma}{E_2} + \frac{\sigma}{\eta_3} t$
- the slope at long times yields the viscosity η_3
- at $t=0$, $\varepsilon = \frac{\sigma}{E_1}$ yielding E_1



Example 1: Velveeta[®] cheese

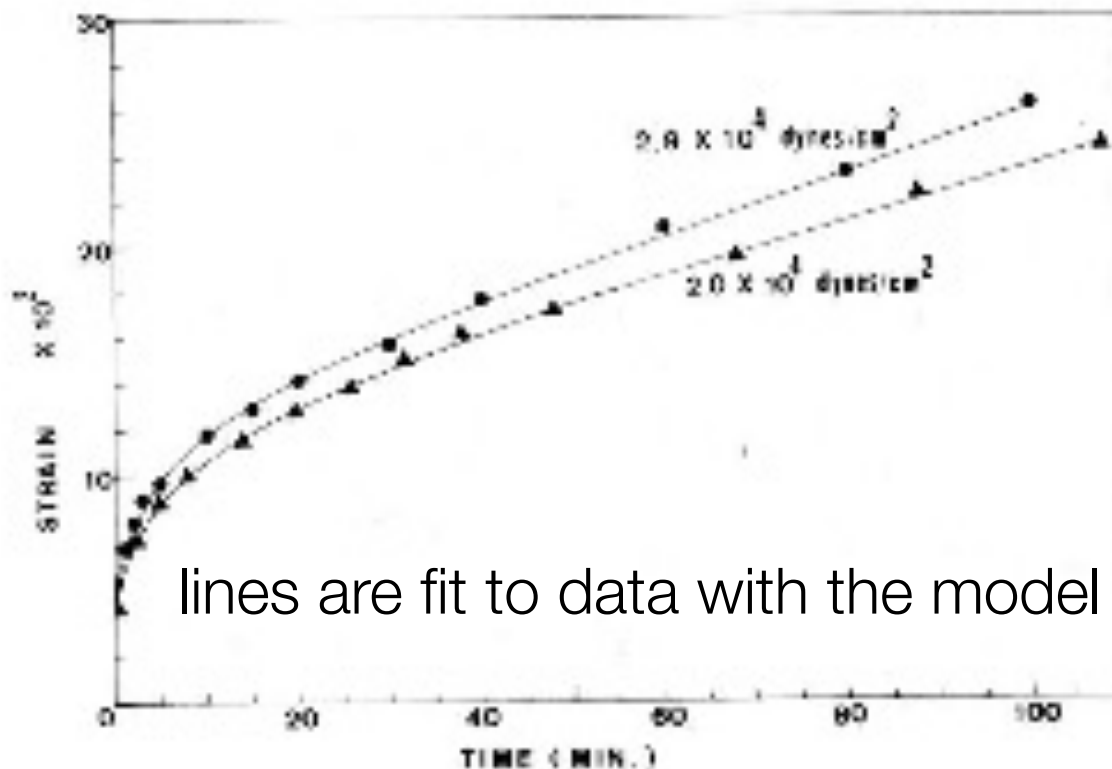
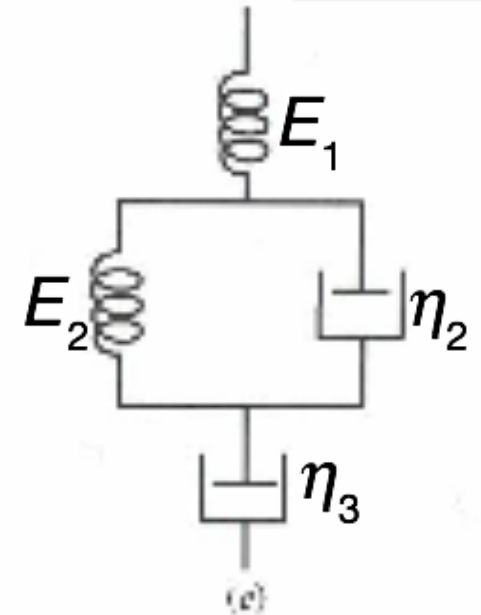
$$\varepsilon = \frac{\sigma}{E_1} + \frac{\sigma}{E_2} \left(1 - e^{-t/\tau_2}\right) + \frac{\sigma}{\eta_3} t$$



- they measured the total strain, and can determine the strain of the spring with modulus E_1 from $\varepsilon_1 = \sigma / E_1$
- the strain of the bottom dashpot is $\varepsilon_3 = \dot{\varepsilon} \Delta t = (\sigma / \eta_3) \Delta t$
- the strain for the Voigt structure is $\varepsilon_2 = \varepsilon - (\varepsilon_1 + \varepsilon_3)$
- E_2 and $\tau = \eta_2 / E_2$ can be found by curve fitting
- thus, have all the parameters

Example 1: Velveeta[®] cheese

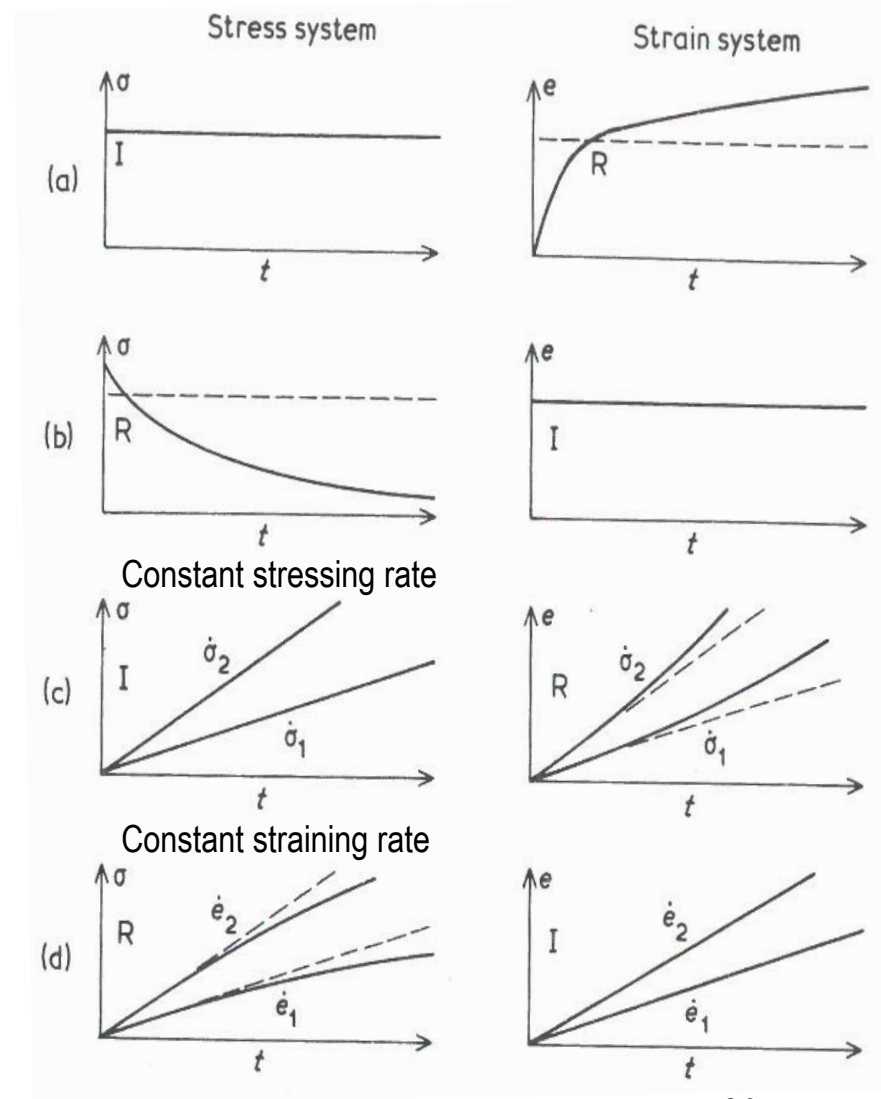
$$\varepsilon = \frac{\sigma}{E_1} + \frac{\sigma}{E_2} \left(1 - e^{-t/\tau_2}\right) + \frac{\sigma}{\eta_3} t$$



Viscoelasticity

- If a constant shear stress, σ_1 , is applied to a viscoelastic material over time, t , strain, ε_1 , evolves as shown in (a)
- Hence at an arbitrary time, t , the strains at the two stresses are $\varepsilon_1(t)$ and $\varepsilon_2(t)$, then;

$$\frac{\varepsilon_1(t)}{\sigma_1} = \frac{\varepsilon_2(t)}{\sigma_2}$$



Viscoelasticity

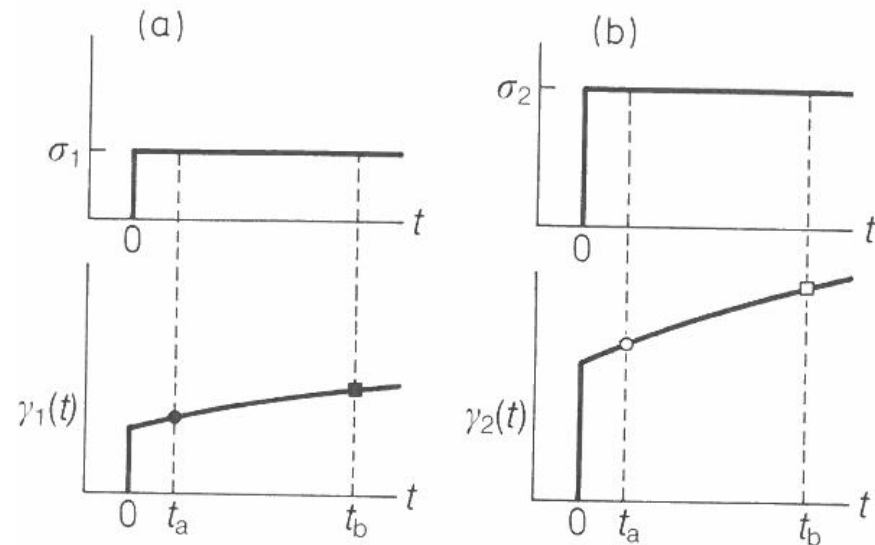
- If a constant shear stress, σ_1 or σ_2 , is applied to a viscoelastic material over time, t , strain, ε_1 (i.e. γ_1) evolves as shown in (a)
- Hence at an arbitrary time, t , the strains at the two stresses are $\varepsilon_1(t)$ and $\varepsilon_2(t)$, then;

$$\frac{\varepsilon_1(t)}{\sigma_1} = \frac{\varepsilon_2(t)}{\sigma_2}$$

- Strain in the two experiments are proportional to imposed stress. Hence creep compliance at time t can be defined as;

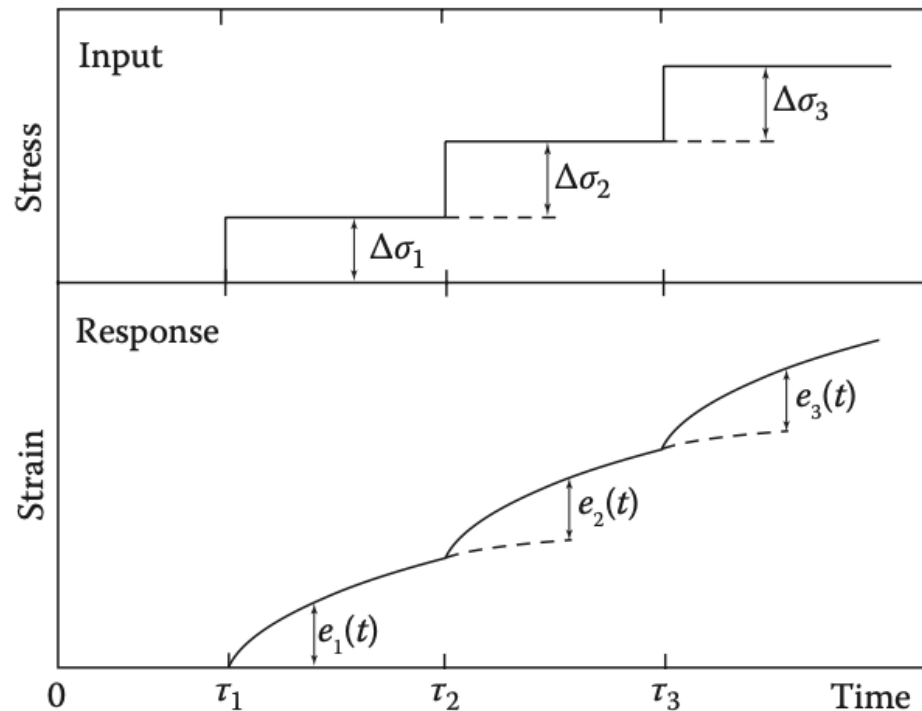
$$J(t) = \frac{\varepsilon_1(t)}{\sigma_1} = \frac{\varepsilon_2(t)}{\sigma_2}$$

in general, $J(t) = \frac{\varepsilon(t)}{\sigma}$



Boltzmann Superposition Principle

- Helps determine **state of stress or strain** in viscoelastic materials from knowledge of the **materials deformation history**
- For varying stress, overall deformation is related to the sum of strains due to each stress step.



Boltzmann Superposition Principle

- Helps determine **state of stress or strain** in viscoelastic materials from knowledge of the **materials deformation history**
- For varying stress, overall deformation is related to the sum of strains due to each stress step.

- We define creep compliance as

$$J(t) = \varepsilon(t)/\sigma, \quad \text{which is only a function of time}$$

then for each loading,

$$\varepsilon(t) = \Delta\sigma J(t - \tau)$$

- With further stressing we get;

$$\varepsilon(t) = \varepsilon_1(t) + \varepsilon_2(t) + \varepsilon_3(t) + \dots$$

$$= \Delta\sigma_1 J(t - \tau_1) + \Delta\sigma_2 J(t - \tau_2) + \dots$$

$$= \sum_{n=0}^n J(t - \tau) \Delta\sigma_n$$

Boltzmann Superposition Principle

- We have seen that overall deformation (strain) is given by;

$$\varepsilon(t) = \sum_{n=0}^n J(t - \tau) \Delta \sigma_n$$

- Which can be represented as an integral ($-\infty$ to t = all deformation history);

$$\varepsilon(t) = \int_{-\infty}^t J(t - \tau) d\sigma(t)$$

- This equation can be express in terms of relaxation time, τ ;

$$\varepsilon(t) = \int_{-\infty}^t J(t - \tau) \frac{d\sigma(\tau)}{d\tau} d\tau$$

Predicts overall strain
After stressing the sample

Stress relaxation & creep

Kelvin Model

- At constant load;

$$\frac{d\varepsilon}{dt} = \frac{\sigma}{\eta} - \frac{E\varepsilon}{\eta}$$

- But if the load is then removed ($\sigma = 0$) after straining the system, then;

$$\frac{d\varepsilon}{dt} = \frac{E\varepsilon}{\eta} \Rightarrow \frac{d\varepsilon}{\varepsilon} = \frac{E}{\eta} dt$$

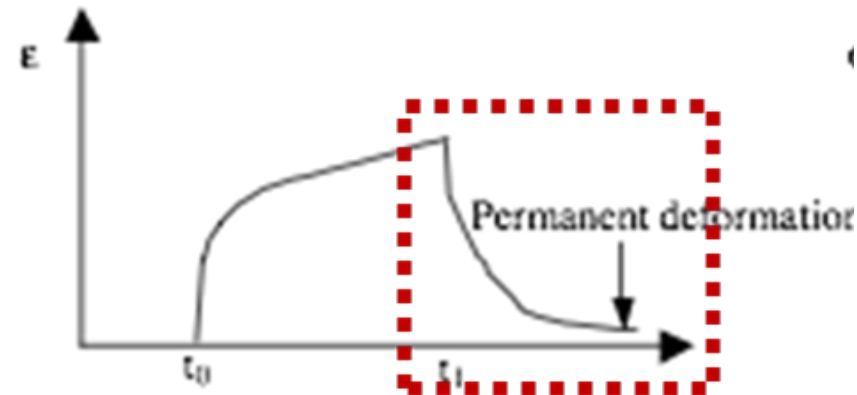
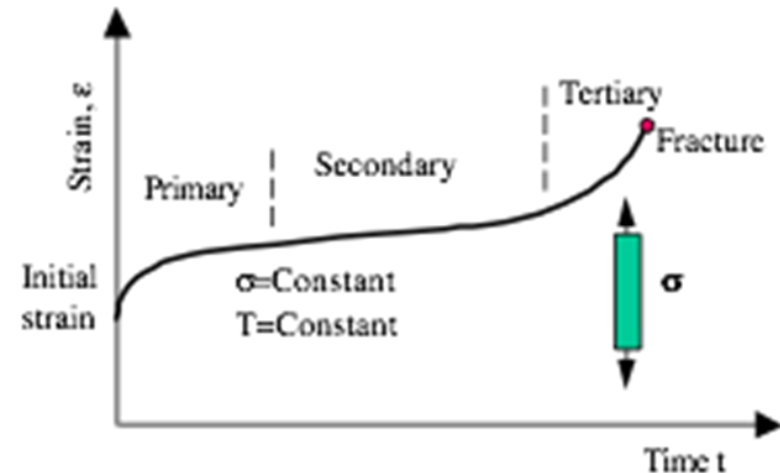
- Therefore;

$$\varepsilon = \varepsilon_0 e^{-Et/\eta}$$

- Since $\tau = \eta/E$; then;

$$\varepsilon = \varepsilon_0 e^{-t/\tau}$$

- Therefore, this model only explains strain relaxation and is useful in explaining creep ([strain relaxation](#))



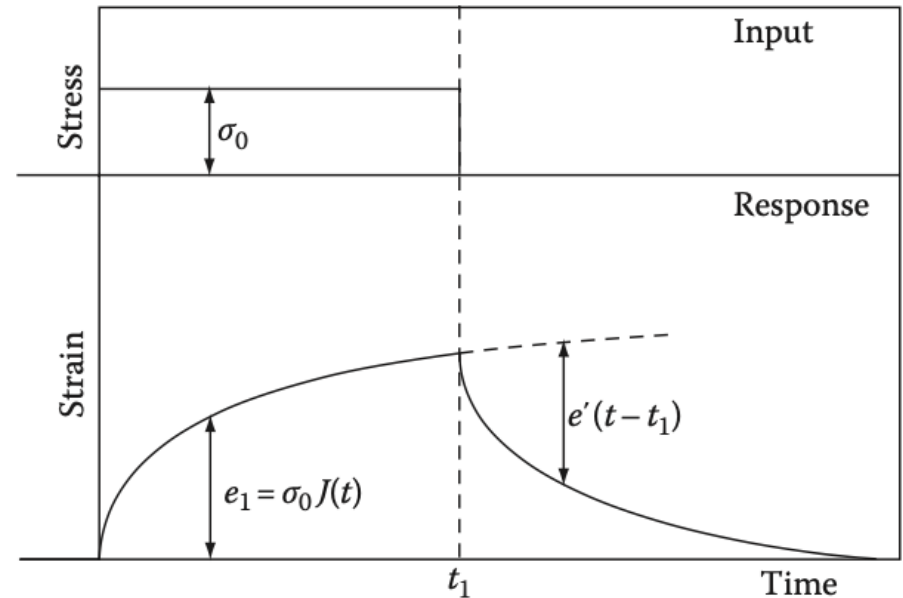
Boltzmann Superposition Principle: Recovery

$$e_1 = \sigma_0 J(t)$$

$$e_2 = -\sigma_0 J(t - t_1)$$

$$e(t) = \sigma_0 J(t) - \sigma_0 J(t - t_1)$$

$$e'(t - t_1) = \sigma_0 J(t - t_1)$$



Boltzmann Superposition Principle

- We can also consider stress relaxation the same way:

$$G(t) = \frac{\sigma(t)}{\varepsilon}$$

- If we consider the stress relaxation modulus, then;

$$\sigma(t) = G(t)\varepsilon = \Delta\varepsilon_1 G(t - \tau_1) + \Delta\varepsilon_2 G(t - \tau_2) + \dots$$

- This is analogous to what we saw above, hence;

$$\sigma(t) = \int_{-\infty}^t G(t - \tau) \frac{d\varepsilon(\tau)}{d\tau} d\tau$$

Predicts overall stress
After straining the sample